M KIRTHIK ADITYA

Cybersecurity & Cloud Security Engineer

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PROFESSIONAL SUMMARY

Cybersecurity engineer with hands-on experience in penetration testing, network security, and vulnerability assessment, backed by Linux, Docker, and AWS. Built a cloud-based cybersecurity lab platform in Go and a verifiable drive-erasure tool in C++ that sanitizes disks at the firmware level and issues signed certificates. Focused on Cybersecurity, Network Security, Cloud Security, and DevSecOps.

PROJECTS

BIT CyberSpace — Cybersecurity & Cloud PaaS Platform

Technologies: Go, Docker, Linux, Nginx, Wireshark, Nmap, Cloud Infrastructure

- Developed backend services in Go to power a cloud-based cybersecurity lab platform, enabling students to run penetration tests against containerized environments and analyze PCAP files to capture flags.
- Hosted 12+ containerized lab environments supporting 20 concurrent users, with purpose-built vulnerable environments for hands-on exploitation practice.
- Built and managed Docker-based containerized environments on Linux, following secure configuration practices to isolate lab workloads from the host and from each other.
- Configured Nginx for reverse proxy, TLS termination, and service routing, cutting external attack surface by 40% and centralizing access control to backend containers.
- Integrated Wireshark- and Nmap-based workflows for traffic capture/analysis and reconnaissance exercises, and designed the architecture around cloud security principles (network segmentation, least-privilege access), reducing environment provisioning time by 80% through deployment automation.

ZeroTrace — Verifiable Drive Erasure, Bootable Linux ISO

Technologies: C++, Linux, NVMe, OpenSSL, Cryptography, Debian live-build

- Built a bootable Debian ISO (UEFI and BIOS) in C++ that erases drives with their own NVMe and ATA sanitize commands, sending raw commands through the kernel's NVMe admin and SG_IO interfaces and detecting 9 sanitize methods per drive.
- Verified every erase by reading ~1,000 regions back with O_DIRECT; for firmware erases it plants random canaries first, so a drive that claims success without erasing gets caught.
- Issued tamper-evident certificates signed with Ed25519 over RFC 8785 canonical JSON, with matching verifiers in C++ and JavaScript and a public browser verifier.
- Found and fixed boot, D-Bus, and console-font issues by testing the ISO in Proxmox VMs, with a destructive integration test that locates its target disks by serial number.

TECHNICAL SKILLS

Programming	Go, C++, Bash Scripting
Cloud	AWS, IAM & Access Control
Containers & DevOps	Docker, Linux, Git/GitHub
Security	Wireshark, Burp Suite, Nmap, Metasploit, Vulnerability Assessment, Penetration Testing
Networking	TCP/IP, DNS, HTTP, OSPF, Packet Analysis

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning

Bannari Amman Institute of Technology | 2024 – 2028

CERTIFICATIONS & AREAS OF INTEREST

- Cisco – Introduction to Cybersecurity
- Pwn Till Dawn — online.pwntilldawn.com/Achievements/9259 — #13 in India, #466 on the battlefield, 1,315 points, 8 machines

Interests: Cloud Security | DevSecOps | Network Security | Cybersecurity Engineering

CTF & SECURITY COMMUNITY ACTIVITY

- Active on tryhackme.com/p/SPRYZZEN — solving rooms and CTF-style challenges across web, network, and cloud security tracks.
- Publish cybersecurity and cloud security write-ups on medium.com/@mkirthikaditya.al24, translating hands-on lab and project work into technical blog posts.
- Hands-on practice in Linux administration, networking, and cloud environments through personal homelabs, with code and configs on github.com/spryzz3n.